GENERAL SPECIFICATION G5.20

Cable Installations

CONTENTS

G5.20 CABLE INSTALLATIONS

G5.20.1 General

(a) This section is supplement to General Specification G5.12 – Conductors, and shall be read in conjunction with the General Specification on Conductors.

(b) Any cable installation requirements not specified in this section but imposed by SP Group or other regulatory authority shall be complied with.

(c) Any cable installation requirements not specified in this section but imposed by MHA on segregation for Cat 1 Critical Infrastructure shall be complied with.

G5.20.2 Electrical Installation

(a) The electrical installation shall be carried out by a Licensed Registered Electrical Contractor in Singapore and the installation shall fully comply with all statutory requirements and the specifications of this Tender Document.

(b) A complete electrical installation for all items of plant shall be carried out by the group of competent and trained workers under the supervision of competent person.

G5.20.3 Standard Cable Installation

(a) Cable termination procedure shall be carried out strictly in accordance with the cable manufacturer's instructions. The Contractor shall fully comply with the safety regulations imposed by SP Group, MOM, EMA or other regulatory authority for the installation of cables.

(b) The numbers of cable draw boxes and ducts shown on the tender drawings are only indicative. The Contractor shall also construct cable ducts and opening in the existing and new buildings and structures. Where the openings for cable ducts and cable draw boxes in the existing and new buildings are not available, the Contractor shall provide these openings and cable ducts with proper drawings and cables routes at an early stage. Opening and cables ducts in the existing structures shall be provided under this Contract.

(c) The Contractor shall be responsible for the boxing out and cutting out of all openings, boxings, etc. required on the existing buildings and structures. All plinths, concrete piers, supports and fixings required for the complete and satisfactory installation of equipment shall be provided.

(d) Cables within the buildings shall be neatly cleated and run on cable trays or in PVC ducts. Control and instrumentation cables, when installed in conduit system, shall not be run with any power cables but shall be run in a separate system.

(e) Where cables pass through structural concrete or brick walls which are either below ground level, or enclose a space that ought to be sealed for another reason (eg. odour, ventilation, fire compartment), they shall be sealed through the wall by the use of Multi-Cable Transit Systems. The Multi-Cable Transit System adopted shall be subject to the approval of the S.O..

(f) The number of cable ducts which will be laid under roads shall be indicated on the shop drawings.

(g) The following minimum bending radius will serve as guidelines for cable installation. The Contractor shall install the cables in accordance with the cable manufacturer's requirements:

(i) Single core cables: 20 x diameter of cable; and

(ii) Multicore cables: 15 x diameter of cable

G5.20.4 Excavation and Reinstatement and Laying of Cables

(a) Cables passing under roadways shall be drawn through HD uPVC ducts which shall be encased in concrete block.

(b) Underground cables shall be laid direct in duct bank, unless otherwise indicated. They shall be drawn through ducts where indicated.

(c) Excavation works for cable installations shall only be carried out when the final ground levels have been attained and no further civil works excavation are envisaged (where applicable).

(d) Trenches shall be excavated to provide the following minimum cover.

(e) In open ground and under pavements: All cables up to and including low voltage 500 mm below ground level.

(f) Under roadways subject to vehicular traffic: All cables up to and including low voltage shall be laid 800 mm below ground level. High voltage cables shall be laid 1100 mm below ground level.

(g) Turf and top soil shall be removed carefully and preserved for reinstatement in their original positions.

(h) Broken land drains and damage to other services shall be reported and indicated. Clearance shall be obtained from the S.O. before backfilling the trench.

(i) The excavation shall be kept free of water and properly shored up. Other services uncovered shall be adequately supported by slings or other means and protected.

(j) Before cables are laid, the bottom of the trench shall be evenly graded, cleared of loose stones and then covered with a 75 mm layer of earth which has been passed through a sieve with a maximum mesh of 10 mm or, where the ground is unsuitable, with sand.

(k) Power cables shall be pulled in over adequately spaced cable rollers. In straight run trenches cable crossing is not permitted except where cables branch from the main run. At each draw in point, joint or junction box, the cable shall be left slack. Cables shall not be pulled taut to straighten them after laying. In order to ensure that the strain is taken on the cores as well as the sheath when cables are laid with a cable stocking, a solid plumbed hauling end shall be made.

(l) When more than one cable is laid in a duct bank the cables shall be spaced apart in accordance with their rating or with other services to the following minimum spacing:

(i) Between high voltage cable and high voltage cable: 50 mm;

(ii) Between high voltage cable and low voltage cable: 300 mm;

(iii) Between high voltage cable and 110 Vdc control cable: 300 mm;

(iv) Between high voltage cable and 24 Vdc control cable: 600 mm;

(v) Between high voltage cable and instrumentation/telephone cable: 600 mm;

(vi) Between low voltage cable and low voltage cable: Touching, unless current considerations dictate otherwise;

(vii) Between low voltage cable and 110 Vdc control cable: 150 mm;

(viii) Between low voltage cable and 24 Vdc control cable: 300 mm;

(ix) Between low voltage cable and instrumentation/telephone cable: 300 mm; and

(x) Between control cable: Touching.

(m) When the minimum segregation distance between cables and services is not possible a 50 mm thick concrete or stone tile shall be used as separator.

(n) After laying, the cables shall be covered with enough fine sieved earth (or sand where the earth is unsuitable) to ensure a 50 mm cover after tamping. Protective cable slab and warning covers as specified herein shall be laid over the cables and the trench filled in and compacted.

(o) Reinstatement shall be effected by back filling in 100 mm layers and hand ramming the first two layers. Power rammers may be used for the remaining layers. Turf shall be replaced and the level of the finished reinstatement shall not protrude more than 25 mm above normal ground level.

G5.20.5 Protective Cable Slabs and Cables Warning Tapes

(a) Reinforced concrete slabs shall be provided as protective cable slabs. They shall be approximately 30 mm thick, 600mm long, 200 mm wide and suitably size for handling by one man.

(b) Protective cable slabs shall be engraved ‘INSTRUMENTATION’, or ‘ELECTRICAL LT’ or ‘CONTROL’ or according to the service. They are to be laid horizontally, interlocking successively one with the other both vertically and laterally. Special cover shall be provided, if required, for short radius bends.

(c) Protective cable slabs and interlocking cable warning tapes shall be provided for cables laid direct in the ground. They shall be of such width as to provide a 50 mm margin of cover on both sides of the cables.

(d) Cable warning tapes of PVC type shall be used. The colour of the cover plates for the HV and LV cables shall be in yellow and black respectively. They shall be continuously marked ‘CAUTION- ELECTRICAL CABLES BELOW’.

G5.20.6 Cable Markers

(a) The location of all buried cables shall be marked by concrete slab markers 600 mm square and 100 mm thick as directed by the S.O..

(b) HV and LV cables. Each cable run shall be marked at the point where it leaves the plinth, sub-station, feeder pillar or other current controlling device and shall be marked at approximately every 50m along the cable run with an additional marker at each change of direction of the cable run.

(c) Cable markers shall be installed flat in the ground immediately above the cable with approximately 25 mm projecting above the surface. The marker shall bear the words `HV cable' or `LV Cable' as required.

(d) The location of each underground cable joint shall be marked by a concrete slab placed over the joint with approximately 25 mm of slab projecting above the ground. The slab shall bear the word `joint'.

G5.20.7 Identification Tags

(a) The Contractor shall provide Identification Tags for all power cables supplied by him. The Contractor shall label both ends of all control and signal cables.

(b) The tags shall be 100 mm by 300 mm stainless steel plates of minimum thickness 1.5 mm. The tags for the cables shall indicate designation, size, type, cables number and rating and the size of tags shall depend on the cable size. Tags shall be provided for all cables and attached to the cables or conduits at entry and exit.

(c) The identification letters and numbers shall have a minimum height of 10 mm and shall be engraved on the plates and highlighted using chloride free paints.

(d) The tags shall be securely fastened to the identified components using brass rings, small solid link brass chains or other approved methods. The tagging shall be completed two weeks before the commissioning of the work.

(e) All cable label shall be standardized as “From/To” format.

G5.20.8 Junctions Boxes and Cable Glands

(a) Junction boxes shall be watertight and be of stainless steel material. Flameproof boxes shall be used in hazardous locations.

(b) Cable glands of British Standard Specifications shall be used. In general, brass glands shall be used. Stainless steel glands shall be used for XLPE/AWA/PVC cables except for LV motor terminations where brass glands may be used together with stainless steel Jubilee clips which are earthed together.

(c) Glands shall be weatherproof, water-tight or flameproof as the location of installation requires. Double-seal cable glands shall be used where the cables are terminated in IP 55 (or better) enclosures.

(d) Glands for armoured cables shall be of two-piece construction without the requirement of a separate earthing ring.

G5.20.9 Cable Ducts

(a) Ducts shall be 100 mm minimum bore unless specified otherwise. They shall be of material, manufacture and method of jointing approved by the S.O. as follows:

(i) Earthenware, either salt glazed or ceramic glazed internally, of the self-aligning type or plastic sleeve type with peeling flange suitably flexible to ensure self-alignment of the bore;

(ii) HD uPVC for all types of cables;

(iii) Fibre with collar joints; and

(iv) Corrugated polyethylene.

(b) Trenches for earthenware ducts shall be scooped out at all points where the sockets rest so that the body of the duct lies upon solid ground. In rocky soils, a layer of loose earth shall be spread over the bottom of the trench and rammed to afford a bedding for the ducts. In waterlogged or unstable soil a foundation for the ducts shall be provided of concrete to the satisfaction of the S.O..

(c) Fibre and PVC ducts shall be laid on and surround with concrete to provide a minimum cover of 150 mm above and below and 100 mm on each side of the collars.

(d) Ducts shall be kept clear of gas and water pipes, drains, sewers and electrical plant. Clearance for gas and water mains shall be at least 150 mm to allow for "tapping" equipment. Clearance shall not be less than 25 mm. Where services cross, the minimum clearance shall be 50 mm.

(e) Ducts shall be laid to provide the minimum cover and spacing specified elsewhere in this section.

(f) Ducts shall be plugged with wooden plugs inserted at the ends of sections to prevent the entry of soil and stones.

(g) Ducts shall be cleared just before any cables are drawn in by dragging a mandrill of diameter 10 mm less than internal duct diameter followed by a circular brush 10 mm greater in diameter than the ducts.

(h) After cables have been installed, the ends of the cables duct shall be sealed with non-shrink grout to form a gas, water and fire barrier. Cables entering the ducts shall be properly supported to prevent ground settlement giving rise to excessive bending to the cable.

(i) Cables occupying narrow ducts may be laid on the bottom. When the duct width exceeds 300 mm cables shall be firmly fixed direct to the duct walls or to a cable tray.

(j) Cable ducts to be located below driveway (such as at road crossing) must be encased in concrete with adequate strength to withstand the maximum dynamic and static load likely to be imposed on top. Maximum vehicle load shall be based on heaviest vehicle and equipment likely to run across it or park on top.

(k) At least 2 spare ducts shall be provided at each road crossing and at each building entry, unless otherwise approved by the S.O..

END OF SECTION